Lenses and image formation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use ray directions to explain converging and diverging lens behavior.

SUCCESS CRITERIA - I can...

1. Distinguish converging and diverging lenses.
2. Trace principal rays with a clear optical axis.
3. State whether an image is real or virtual in the given case.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “lens” and “focal point” to explain your answer. Sentence starters are optional.

lens: a transparent element that changes light-ray directions

focal point: where suitable parallel rays meet or appear to originate

ray: a line showing the modeled direction light travels

01 Warm-up: bring back an earlier idea

A source approaching a stationary listener at **15 m/s** emits **500 Hz**.
1. Use v_{sound}=**340** and $f_{obs}=f_s\ v/(v-v_s)$.
2. **Find observed frequency**.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

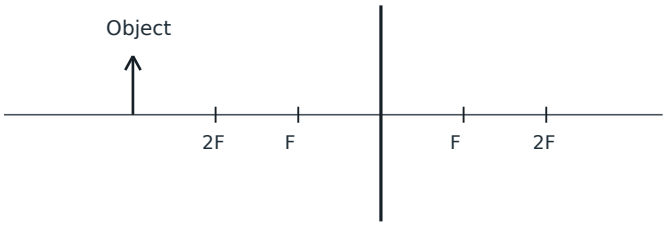
02 Worked example: explain the prediction

Reference: $\mathbf{1}/f = \mathbf{1}/d_o + \mathbf{1}/d_i$.
Thin lens: f is focal length; o = object, i = image. Use the stated sign convention and a ray diagram.

Worked reasoning: A converging lens can form a real image of an object beyond its focal point. A ray parallel to the axis passes through the far focus; a central ray is nearly straight when the lens is treated as very thin.

Your turn: Draw these two rays and **mark their intersection** for an object beyond **2** f.

Sentence starter: In my work, I used … to show … . I labeled it … .



Converging thin lens; add principal rays.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Complete ray sketches for parallel rays through a converging lens and a diverging lens. **Label the optical axis 0**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Left: converging lens. Right: diverging lens. Horizontal lines mark the optical axes.

Marzano check - circle: ① ② ③

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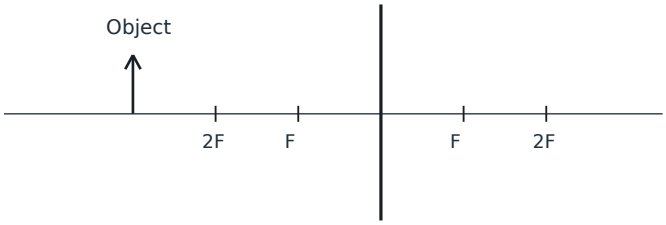
Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A converging lens has focal length **10 cm**. A small lamp is **30 cm** from it.
1. **Label the diagram**, find image distance using **1**/f = **1**/d_o + **1**/d_i, and **calculate m** = -d_i/d_o.
2. **Check the sign** and size with two principal rays.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Converging thin lens; add principal rays.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A magnifying glass produces an upright virtual image when object is inside f. **Explain why** “converging lens always makes a real image” is wrong.

Sentence starter: The step I would change is ... . I would change it because I found ... .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A wave snapshot has crests at **0.2**, **0.6**, **1 m**. **Find wavelength**. Its period is **0.06 s**; **find frequency**. **Label a wave sketch** to **support the calculation or comparison**.

Word help: wavelength = distance between corresponding points on repeated waves

Sentence starter: I found ... [unit]. This value means ... in this situation.



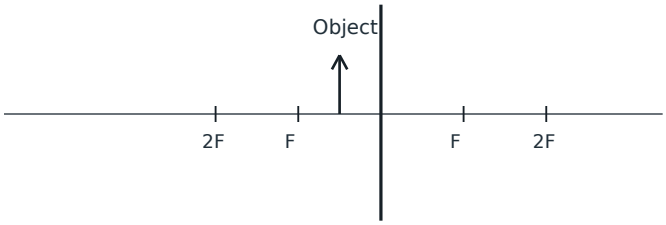
Time or position: choose and label

Marzano check - circle: ① ② ③

07 Exit check

For a converging lens f=**10 cm** and d_o=**5 cm**, find d_i and m. **State image kind**.

Sentence starter: I found ... [unit]. This value means ... in this situation.



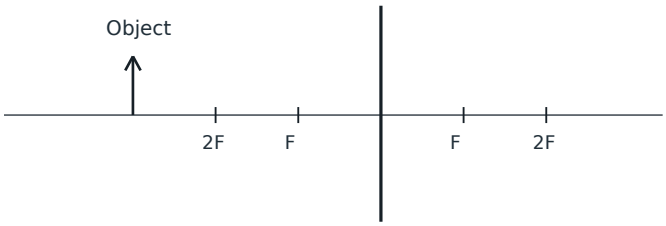
Converging thin lens; add principal rays.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A diverging lens has f = **-10 cm** and an object **30 cm** in front.
1. Use **1**/f = **1**/d_o + **1**/d_i and m = -d_i/d_o.
2. **Find image distance** and magnification, then **check them** with a ray sketch.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Diverging thin lens; add principal rays.

Marzano check - circle: ① ② ③